even more, it’s not even well defined: that’s the price to pay to speed up lookup by value.

To really understand how efficiently we can solve this problem, we need to get into the details of implementations. Using the dictionary abstraction, however, allows us to discuss how to solve a problem (for instance finding if an email belongs to a list of contacts) without having to deal with the details of the data structure’s representation, and hence leaving us free to focus on the task itself.

4.3 Describing the Data Structure API: Associative Array

An associative array (also referred to as *dictionary*[2], *symbol table*, or just *map*), is composed of a collection of (key, value) pairs, such that:

1. Each possible key appears at most once in the collection;
2. Each value can be retrieved directly through the corresponding key.

The easiest way to grasp the essence of associative arrays is thinking about regular arrays as a special case: the keys are just the set of indices between `0` and the size of the array minus `1`, and we can always retrieve a value by providing its index, so that the (plain) array `[“the”, “lazy”, “fox”]`